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|----------------|--------------------|----------------|--|--|--|--|--|--|
| EXPT.<br>NO. 1 | NAME<br>slit width | M T W T F S S  |  |  |  |  |  |  |
|                |                    | Page No.: 01   |  |  |  |  |  |  |
|                |                    | Date: 18/09/24 |  |  |  |  |  |  |
|                |                    | YOUVA          |  |  |  |  |  |  |

◦ Aim:

- To calculate the slit width of a single slit and a blade using He-Ne laser and Travelling Microscope.

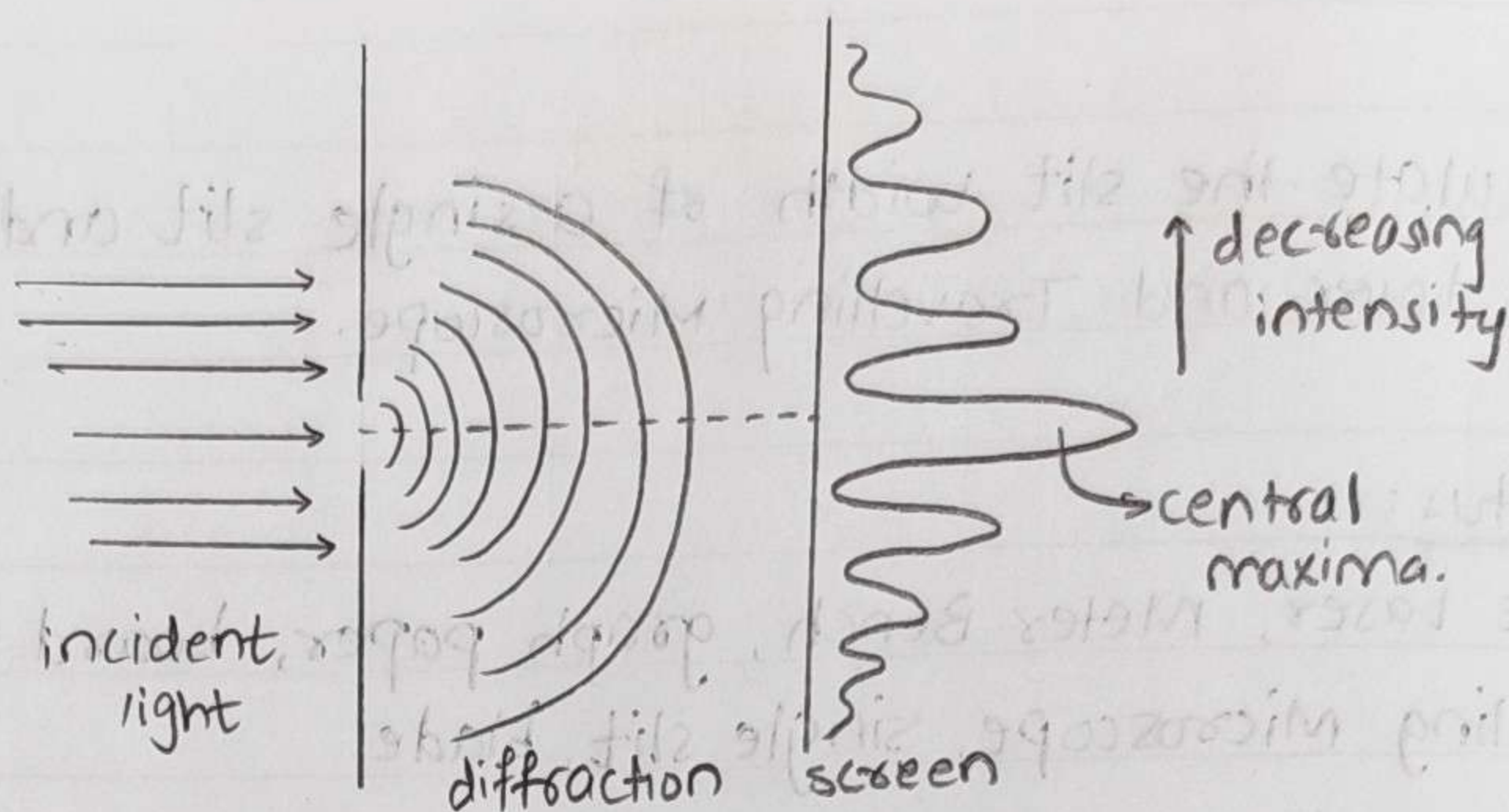
◦ Apparatus:

- He-Ne Laser, Meter Bench, graph paper, board (stand/screen), Travelling Microscope, single slit, blade.

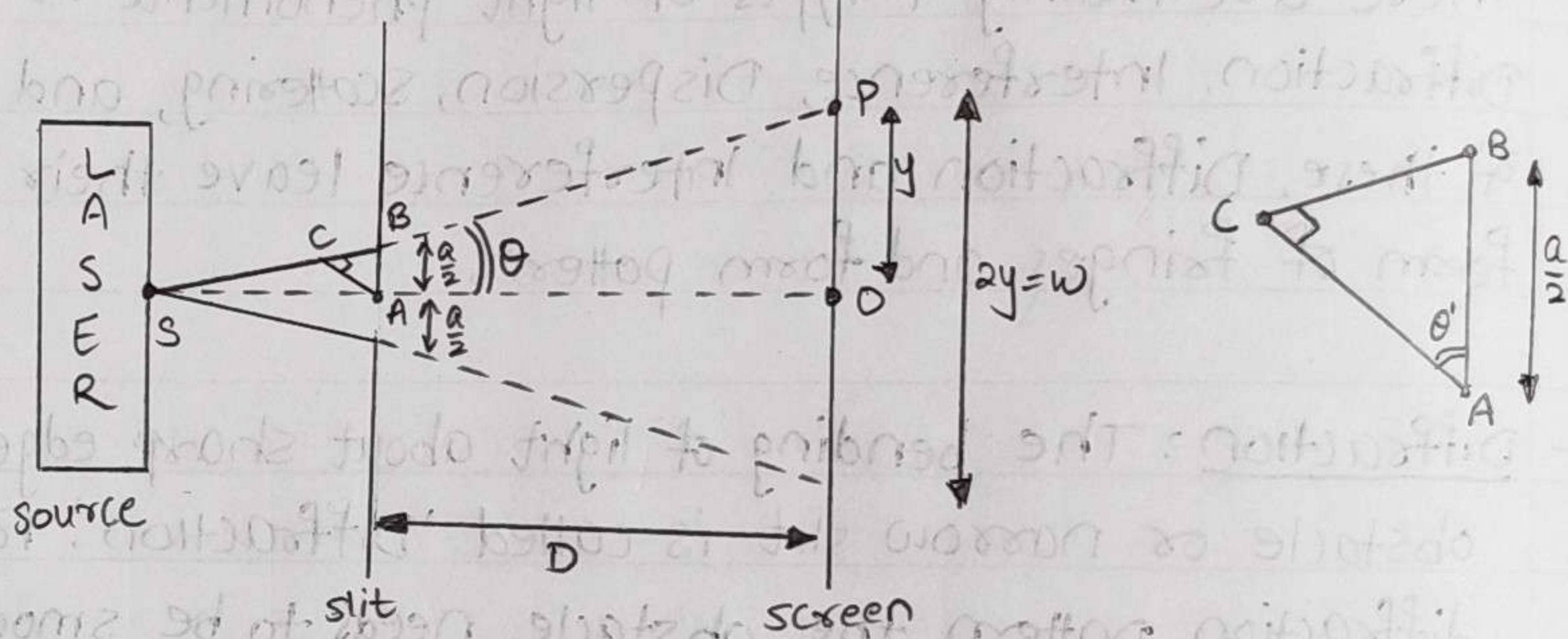
◦ Theory:

- There are mainly 7 types of light phenomena viz. Reflection, Refraction, Diffraction, Interference, Dispersion, Scattering, and Polarization. Out of these, Diffraction and Interference leave their occurrence in the form of fringes and form patterns.
- Diffraction: The bending of light about sharp edges of an opaque obstacle or narrow slit is called 'Diffraction'. For getting a proper diffraction pattern, the obstacle needs to be smooth.
- There are mainly two types of diffraction:
  - (i) Fresnel Diffraction: In Fresnel diffraction, the source and screen are at a finite distance from the obstacle and the incident light wavefront is either spherical or cylindrical.
  - (ii) Fraunhofer Diffraction: In Fraunhofer diffraction, the source and screen are at infinite distance from the obstacle and the incident light wavefront is planar.

# Figures:



[Figure (1)]



[Figure (2)]

- Here, we are performing Fraunhofer diffraction.

The light coming from infinite distance falls on the aperture and undergoes diffraction. The diffraction pattern is copied to the screen and form fringes such that a central maxima is of maximum intensity, and then minima, maxima alternates with diminishing intensity, as shown in figure (1).

$$\therefore I = \left( \frac{\sin \beta}{\beta} \right)^2 I_0 \quad \text{where } \beta = \frac{\pi a \sin \psi}{\lambda}$$

where  $\psi$  is the wave function

$a$  is the slit width

$\lambda$  is the wavelength of light.

Differentiating w.r.t.  $\beta$  to get maxima,

$$\therefore \sin \beta (\sin \beta - \beta \cos \beta) = 0$$

$$\Rightarrow \sin \beta = 0 \Rightarrow \beta = 0, \pi, 2\pi, \dots, n\pi$$

$$\therefore n\pi = \frac{\pi a \sin \psi}{\lambda} \Rightarrow \boxed{\sin \psi = \frac{n\lambda}{a}}$$

- Consider figure (2).

$$\text{In } \triangle ABC, \sin \theta' = \frac{BC}{AB} \Rightarrow BC = AB \sin \theta' \quad \text{--- (1)}$$

Now, path difference,  $BC = \delta$

and incident diffraction angle  $\theta' = \theta$

As  $\theta$  is very small angle,  $\theta \approx \sin \theta \approx \tan \theta$

$$\text{In } \triangle AOP, \tan \theta' = \frac{OP}{AO} = \frac{y}{D} \Rightarrow \theta = \frac{y}{D}$$

$$\text{From (1), } BC = AB \sin \theta' \Rightarrow \delta = \frac{a}{2} \theta$$

$$\text{Phase difference, } \phi = \frac{2\pi \delta}{\lambda} = \frac{2\pi}{\lambda} \left( \frac{a y}{2 D} \right) \Rightarrow \phi = \frac{\pi a y}{\lambda D}$$

• observation Table:

(i) slit width of single slit using Laser: ( $\lambda = 6328 \text{ \AA}$ )

| Sr. No. | central fringe width (w) | Distance (D) | slit width (a) |
|---------|--------------------------|--------------|----------------|
| 1       | 6 mm                     | 400 mm       | 0.084 mm       |
| 2       | 3.5 mm                   | 350 mm       | 0.126 mm       |
| 3       | 3 mm                     | 300 mm       | 0.126 mm       |

$$\therefore (a_{ss})_{avg} = 0.112 \text{ mm}$$

(ii) slit width of blade using He-Ne Laser:

| Sr. No. | central fringe width (w) | Distance (D) | slit width (a) |
|---------|--------------------------|--------------|----------------|
| 1       | 1.1 mm                   | 300 mm       | 0.345 mm       |
| 2       | 1.8 mm                   | 350 mm       | 0.246 mm       |
| 3       | 2.5 mm                   | 400 mm       | 0.202 mm       |

$$\therefore (a_b)_{avg} = 0.264 \text{ mm}$$

(iii) Verifying single slit using Travelling Microscope: ( $LC = 0.01 \text{ mm}$ )

| Sr. No. | MSR (cm) | VSR | TR (mm) |                          |
|---------|----------|-----|---------|--------------------------|
| 1       | 4.1      | 3   | 41.03   | $a_1 = 41.03 \text{ mm}$ |
| 2       | 4.2      | 4   | 42.04   | $a_2 = 42.04 \text{ mm}$ |

$$\therefore \text{Thickness, } a_{ss} = a_2 - a_1 = 0.11 \text{ mm}$$

(iv) Verifying blade using Travelling Microscope:

| Sr. No. | MSR (cm) | VSR | TR (mm) |                          |
|---------|----------|-----|---------|--------------------------|
| 1       | 3.25     | 3   | 3.253   | $a_1 = 32.53 \text{ mm}$ |
| 2       | 3.23     | 2   | 3.232   | $a_2 = 32.32 \text{ mm}$ |

$$\therefore \text{Thickness, } a_b = a_2 - a_1 = 0.21 \text{ mm}$$

For minima,  $\Phi = \pi \Rightarrow \pi = \frac{\pi a y}{\lambda D} \Rightarrow \boxed{a = \frac{\lambda D}{y} = \frac{2\lambda D}{w}} \quad -(2y = w)$

### o Procedure:

- Arrange the slit, screen and source in such a way that laser focusing on the slit and fringes can be seen on the screen.
- Place a graph paper onto the screen so that we can copy the diffraction pattern (dark/bright fringes).
- Measure the width of central bright fringe ( $w$ ) and the distance between screen & aperture ( $D$ ).
- Note down the readings by changing the values of  $D$ , and take three readings. Calculate the value of slit width ( $a$ ) using formula.
- Verify the slit width by using a travelling microscope.
- Repeat the same experiment by taking blade instead of single slit.

### o Calculations:

(1) Slit width of single slit using laser:

$$a_1 = \frac{2\lambda D_1}{w_1} = \frac{(2)(6328 \times 10^{-10})(400 \times 10^{-3})}{(6 \times 10^{-3})} = 0.084 \times 10^{-3} \text{ m} = 0.084 \text{ mm.}$$

$$a_2 = \frac{2\lambda D_2}{w_2} = \frac{(2)(6328 \times 10^{-10})(350 \times 10^{-3})}{(3.5 \times 10^{-3})} = 0.126 \times 10^{-3} \text{ m} = 0.126 \text{ mm}$$

$$a_3 = \frac{2\lambda D_3}{w_3} = \frac{(2)(6328 \times 10^{-10})(300 \times 10^{-3})}{(3 \times 10^{-3})} = 0.126 \times 10^{-3} \text{ m} = 0.126 \text{ mm}$$

$$\therefore (a_{ss})_{avg} = \frac{0.084 + 0.126 + 0.126}{3} = 0.112 \text{ mm.}$$

(ii) slit width of blade using laser:

$$a_1 = \frac{2\lambda D_1}{w_1} = \frac{(2)(6328 \times 10^{-10})(300 \times 10^{-3})}{(1.1 \times 10^{-3})} = 0.345 \times 10^{-3} \text{ m} = 0.345 \text{ mm}$$

$$a_2 = \frac{2\lambda D_2}{w_2} = \frac{(2)(6328 \times 10^{-10})(350 \times 10^{-3})}{(1.8 \times 10^{-3})} = 0.246 \times 10^{-3} \text{ m} = 0.246 \text{ mm}$$

$$a_3 = \frac{2\lambda D_3}{w_3} = \frac{(2)(6328 \times 10^{-10})(400 \times 10^{-3})}{(2.5 \times 10^{-3})} = 0.202 \times 10^{-3} \text{ m} = 0.202 \text{ mm}$$

$$\therefore (a_b)_{avg} = \frac{0.345 + 0.246 + 0.202}{3} = 0.264 \text{ mm.}$$

### Precautions:

- Do NOT focus the laser beam onto your eye as it may cause permanent eye damage.

### Result:

- slit width of single slit using laser = 0.112 mm
- slit width of single slit using Travelling Microscope = 0.11 mm.
- slit width of blade using laser = 0.264 mm
- slit width of blade using Travelling Microscope = 0.21 mm.

- Conclusion: Proved Fraunhofer Diffraction and calculated slit widths of single slit and blade.